

Since $\vec{F}_i = m_i \ddot{\vec{r}}_i$

$$\underbrace{\vec{r}_1 \times (m_1 \ddot{\vec{r}}_1) + \dots + \vec{r}_n \times (m_n \ddot{\vec{r}}_n)}_0 + \sum m_i (\vec{r}_i \times \vec{r}_i) = 0$$

recall $\frac{d}{dt}(\vec{a} \times \vec{b}) = \dot{\vec{a}} \times \vec{b} + \vec{a} \times \dot{\vec{b}}$

$$\frac{d}{dt} [m_1 \vec{r}_1 \times \dot{\vec{r}}_1 + \dots + m_n \vec{r}_n \times \dot{\vec{r}}_n] = 0$$

$\frac{d}{dt}$ of this is the top equation

integrate

$$m_1 \vec{r}_1 \times \dot{\vec{r}}_1 + \dots + m_n \vec{r}_n \times \dot{\vec{r}}_n = \vec{C}_3$$

$$= \sum m_i \vec{r}_i \times \dot{\vec{r}}_i = \sum m_i \vec{r}_i \times \vec{v}_i = \vec{C}_3 \rightarrow \text{Total angular momentum is conserved (constant)}$$

Three constants of integration ($\vec{C}_3$) from the initial conditions $\vec{r}_i$ and $\dot{\vec{r}}_i$

Energy

Now consider the total energy of the system.

$$E = T + U$$

$\uparrow \quad \uparrow$
Kinetic potential

$\vec{r}^T \vec{r}$ (assuming column vectors)

Kinetic energy: $T = \frac{1}{2} \sum m_i v_i^2 = \frac{1}{2} \sum m_i \vec{r}_i \cdot \dot{\vec{r}}_i$

Potential energy: $U = -G \frac{m_1 m_2}{r}$

$\Delta PE = -W = \int F dr \leftarrow \text{gravity is conservative force, so } W \text{ is independent on the path taken}$

Aside: $V = -\frac{1}{2} G \sum_{i=1}^N \sum_{j=1, j \neq i}^N \frac{m_i m_j}{r_{ij}}$ (For N -bodies)

↑
potential function $\vec{F}_i = -\nabla_i V = -\frac{dV}{d\vec{r}_i}$

Now, $\vec{F}_i = m_i \ddot{\vec{r}}_i$

$\sum \dot{\vec{r}}_i \cdot \vec{F}_i = \sum m_i \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i$ dot $\vec{r}$ on both sides

$\sum \dot{\vec{r}}_i \cdot (-\nabla V) = \sum \frac{d}{dt} \left[\frac{1}{2} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right] \rightarrow \frac{1}{2} m_i \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i + \frac{1}{2} m_i \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i$

$-\underbrace{\sum \frac{d\vec{r}_i}{dt} \cdot \frac{dV}{d\vec{r}_i}}_{=-\dot{V}} = \underbrace{\frac{d}{dt} \sum \left[\frac{1}{2} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right]}_{=\dot{T}}$

so, $-\dot{V} = \dot{T} \rightarrow \dot{T} + \dot{V} = 0$

integrate...

$T + V = C_4$

Total energy is conserved

In summary, we have 10 constants of integration

Total linear momentum: $\left. \begin{aligned} \dot{\vec{r}}_{cm} &= \vec{C}_1 \\ \vec{r}_{cm} &= \vec{C}_1 t + \vec{C}_2 \end{aligned} \right\} \vec{C}_1 \text{ and } \vec{C}_2 = 6 \text{ constants}$

Total momentum: $\sum (m_i \vec{r}_i \times \dot{\vec{r}}_i) = \vec{C}_3 \rightarrow C_3 = 3 \text{ constants}$

Total energy: $E = T + V = C_4 \rightarrow C_4 = 1 \text{ constant}$

There are $6N$ equations of motion for the N -body problem

For $N=2$ $6n=12 > 10$ so what?

Remember: General analytical solution does not exist

↳ But some special cases do have analytical solutions

↳ We can simulate/propagate the EDM from I.C.s

↳ Constants of motion must be satisfied at all times

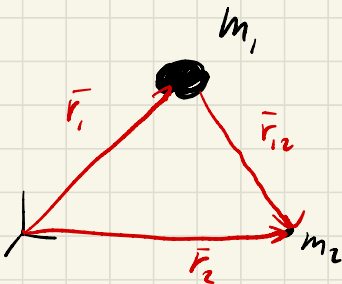
↳ Allowable motion is constrained

Two body problem

Suppose we have a system of only two bodies and wish to describe the motion of one body relative to the other.

Assume:

1. Only two bodies
2. Only gravitational forces
3. Spherical mass distributions



$$\vec{F}_1 = \cancel{m_1} \ddot{\vec{r}}_1 = G \frac{\cancel{m_1} m_2}{r_{12}^3} \vec{r}_{12}$$

$$\vec{F}_2 = \cancel{m_2} \ddot{\vec{r}}_2 = G \frac{m_1 \cancel{m_2}}{r_{12}^3} (-\vec{r}_{12})$$

$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

$$\ddot{\vec{r}} = \ddot{\vec{r}}_2 - \ddot{\vec{r}}_1$$

$$\ddot{\vec{r}}_{12} = \ddot{\vec{r}}_2 - \ddot{\vec{r}}_1 = -\frac{Gm_1}{r_{12}^3} \vec{r}_{12} - \frac{Gm_2}{r_{12}^3} \vec{r}_{12} = -\frac{G(m_1+m_2)}{r_{12}^3} \vec{r}_{12}$$

Typically $m_1 \gg m_2$ Thus, $\vec{r}_{12} \approx -\frac{Gm_1}{r_{12}^3} \vec{r}_{12}$ $Gm_1 = \mu$

Since there are only two bodies... $\vec{r}_{12} = \vec{r}$

Thus, $\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}$ or $\ddot{\vec{r}} = -\frac{\mu}{r^2} \hat{r}$ $\hat{r} = \frac{\vec{r}}{r}$ $r = \|\vec{r}\|$

Specific energy

The specific energy is the total energy divided by the mass

$$\mathcal{E} = \frac{K.E. + P.E.}{m_2} = \frac{T + V}{m_2} \quad T = \frac{1}{2} m_2 v^2 \quad V = -\frac{Gm_1 m_2}{r}$$

$$\mathcal{E} = \frac{1}{m_2} \left[\frac{1}{2} m_2 v^2 - \frac{Gm_1 m_2}{r} \right] = \frac{v^2}{2} - \frac{\mu}{r}$$

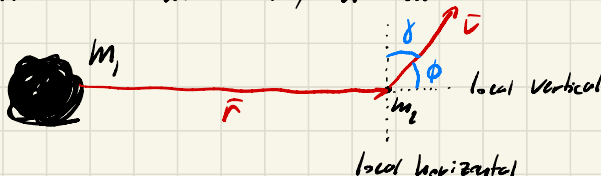
$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} \quad \text{Vis-viva equation}$$

$$\frac{d\mathcal{E}}{dt} = 0 \quad \text{to show conserved}$$

Specific Angular momentum

Recall that angular momentum is

$$\vec{H} = \vec{r} \times (m\vec{v}) = \vec{r} \times m\vec{v} \quad \text{so, } \vec{h} = \frac{\vec{H}}{m} = \vec{r} \times \vec{v} = \vec{r} \times \vec{v}$$



δ is flight path angle
 ϕ is zenith angle

Generally allows us to relate the magnitude of $\vec{h}$ to $\vec{r}$ and $\vec{v}$

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta_{ab}$$

$$\begin{aligned} h = \|\vec{r} \times \vec{v}\| &= r v \sin \phi \\ &= r v \cos \gamma \end{aligned}$$

Review on Kinematics

Translational:

$$\begin{aligned}v &= v_0 + at \\x &= x_0 + vt + \frac{1}{2}at^2 \\v^2 &= v_0^2 + 2a(x - x_0) \\x &= x_0 + \frac{1}{2}(v + v_0)t\end{aligned}$$

Rotational:

$$\begin{aligned}\omega &= \omega_0 + \alpha t \\\theta &= \theta_0 + \omega t + \frac{1}{2}\alpha t^2 \\\omega^2 &= \omega_0^2 + 2\alpha(\theta - \theta_0) \\\theta &= \theta_0 + \frac{1}{2}(\omega + \omega_0)t\end{aligned}$$

$$\begin{aligned}\vec{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\\vec{v} = \dot{\vec{r}} &= \dot{x}\hat{i} + \dot{y}\hat{j} + \dot{z}\hat{k} \\\vec{a} = \dot{\vec{v}} = \ddot{\vec{r}} &= \ddot{x}\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k}\end{aligned}$$

$$\begin{aligned}\vec{r} &= r\hat{e}_r \\\vec{\omega} = \dot{\vec{r}} &= \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta \\\vec{a} = \dot{\vec{\omega}} = \ddot{\vec{r}} &= [\ddot{r} - r\dot{\theta}^2]\hat{e}_r + [2\dot{r}\dot{\theta} + r\ddot{\theta}]\hat{e}_\theta\end{aligned}$$